Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/16/2022
PI Name:	Liu, Bin Ph.D.		
Project Title:	Three-Dimensional Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	07/25/2017	**End Date:**	10/26/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/26/2021 and contract number changed to JPL 1661767 per U. Israelsson/JPL (Ed., 5/24/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goree, John Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1661767; JPL 1579454		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A flight experiment will be performed to study phenomena resulting from shear motion in a dusty plasma. Applying an external shear, we will probe fundamental material properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. We will take a microscopic approach to uncover the nature of macroscopic properties. We will use mesoscopic particles, which serve as analogs of molecules, but allow tracking of individual particles as they interact. The experiment will use dusty plasmas, also known as "complex plasmas." These are four-component mixtures of micron-size solid particles, electrons, ions, and neutral gas. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. Microgravity enables our experiment with a three-dimensional sample. Without microgravity, under 1g conditions, sedimentation causes the dust cloud to settle into a 2D layer, or at most a few horizontal layers. Our microgravity experiment will extend the understanding of particle motion that was gained in previous ground-based experiments with smaller 2D samples. Our flight experiment will be performed using Plasma Kristall–4 (PK-4), which is a facility in the Columbus module of the International Space Station (ISS). It was built by the European Space Agency. The main diagnostic is video imaging using cameras. Image data will be analyzed to generate particle position and velocity. Physical measures that can be obtained from these image data, using methods developed for ground-based experiments, include a wave's propagation speed and dispersion relation, temperature profile, flow velocity, or shear stress. SCIENTIFIC/TECHNICAL MERITS The experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? The experiment will exploit manipulation devices on PK-4 including laser manipulation. Laser light imparts momentum to the solid particles, causing them to flow through the surrounding sample region. This modulation will be a pulse that is either long or short, according to the scientific goal. The feasibility of our flight experiment is based on previous ground-based experiments and our simulation. Our laboratory experiments have confirmed localized viscous heating and lattice waves. Ground-based experiments by other scientists have also confirmed the existences of plastic flow conditions and stick-slip motion. Using a simulation under PK-4 conditions, we have established the physical mechanism of wave excitation by a pulsed shear flow. In the unlikely event that this flight experiment does not yield useful results, we have a plan for a backup experiment to investigate the superheating and supercooling phase transitions, or non-Gaussian statistics. BROADER IMPACTS This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Task Progress:	Note: Below is the April, 2022 update of project progress and accomplishments, submitted by the PI (Ed., 11/14/22) Project objectives: • Understanding fundamental physics of particle motion in the presence of flows • Investigating statistical physics and transport at a particle level using 3D dusty plasmas Main achievements: • ISS flight experiment We successfully performed a flight experiment in PK-4 Science Campaign 7, using the PK-4 instrument on the International Space Station (ISS). This flight experiment resulted in a peer-reviewed journal paper, titled "Time-Dependent Shear Motion in a Strongly-Coupled Dusty Plasma in PK-4 on the International Space Station (ISS)". Physics findings: 1. Motion in the presence of particle flows. In a three-dimensional (3D) strongly-coupled dusty plasma under microgravity conditions, a stream of particles was momentarily pushed to move through the surrounding sample region. Time-dependent shear motion was investigated in the resulting impulsive unsteady flow. At the onset of the impulsive motion, the flow was observed as it grew in space and time. Later, during the steady manipulation, we find the thickness of the flow region fluctuated significantly over time. 2. Non-Gaussian statistics. The velocity distribution function (VDF) in a 3D dust cloud in PK-4 under microgravity conditions was found to have a non-Maxwellian shape with high-energy tails; it is fit well by a combination of low-energy Maxwellian core and a high-energy non-Gaussian Kappa-distribution halo. Similar non-Maxwellian VDFs are typically observed in space plasmas. 3. Diffusive motion in 3D dusty plasma. Random motion of small solid particles in a 3D dust cluster under microgravity conditions was found to be nearly diffusive. Publications and presentations (see cumulative bibliography for additional detail). • Eight peer-reviewed journal papers o PK-4 ISS flight experiment: 5 o Ground-based experiments: 2 o Simulation: 1 • Fifteen talks at various conferences Recent update: During the six-month-extension period, we published a paper in a peer-reviewed journal, reporting the

	results from our flight experiment performed in PK-4 Science Campaign 7 on the ISS. Highlights from this paper include: (1) an observation of impulsive shear flows in a 3D strongly-coupled plasma; (2) flow fluctuations at different time and length scales; (3) a microscopic approach at atomic-like particle level.
Bibliography Type:	Description: (Last Updated: 10/05/2022)